

Reece Shiraishi

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EDUCATION

Cornell University, College of Engineering (GPA: 3.8)

B.S. Electrical and Computer Engineering, Minors in Computer Science & Artificial Intelligence Expected: May 2027

- **Relevant Courses:** Robotics, Embedded Systems, Computer Architecture, Statistics, Foundations of ML, Digital Logic, Computer Architecture, Microelectronics
- **Achievements:** National Merit Semifinalist | Presidential Scholar Nominee | Eagle Scout | IEEE Executive Board

SKILLS

Programming Languages: C, C++, Typescript, Verilog, Quartus FPGA tools, Python, Bash scripting, R, Arduino, Java, Assembly (ISAs: RISC-V, ARM)

Tools, Libraries, Frameworks: Git/GitHub, ROS/RViz, LTSpice, oscilloscopes, NumPy, PyTorch, Pandas, Scikit-learn, SLURM cluster, CMake, I2C, LTspice, NoSQL databases

Languages: English (native), Spanish (full professional fluency)

EXPERIENCE

ML Engineer Intern | SRC, Inc; Syracuse, NY

May 2026 – Aug. 2026

- Engineered high-throughput workflows in Python for EM signal challenges, communicating results to stakeholders
- Trained and fine-tuned image-recognition model on radar pulses on distributed Linux Slurm cluster
- Developed automated bounding-box testing software for Linux environment, boosting training set integrity

Software Engineering Intern | Fermi National Accelerator Laboratory; Batavia, IL

Jun. 2025 – Aug. 2025

- Developed Python workflow to reformat data (YAML to LaTeX/Markdown); awarded academic citation. System actively used by 7-person research team
- Trained PyTorch neural network to classify particles to 70% accuracy while displaying accuracy, AUC, confusion matrix

Backend Developer Intern | Wabi Skincare; Hoboken, NJ

Jan. 2025 – May 2025

- Architected scalable user engagement features (saved post tracking) in fast-paced startup environment
- Retrieved and stored skincare data and constants in Firebase, improving codebase maintainability
- Resolved bug where website tabs failed to render using asynchronous database retrieval

Research Assistant – Prof. Cafer Avcı | Cornell University; Ithaca, NY

Jan. 2025 – May 2025

- Learned and applied GeoJSON Python packages and OpenStreetMap APIs to improve traffic flow in urban areas
- Consolidated the fragmented Python analysis pipeline, turning 9 programs into performant command-line utility

PROJECTS

CNet Framework / CAST Autograd (C++, CMake)

Aug. 2025 – present

- Architected concurrently programmed deep learning frameworks, including full documentation, from scratch
- Implemented tensor operations, branching structure for arbitrary network architectures
- Leveraged Github Workflows CI/CD for automated testing, ensuring correctness

Theft Protection IoT Device (Embedded C, Assembly)

Apr. 2025 – May 2025

- Developed bare-metal security device for ARM Cortex FRDM-KL46Z microcontroller under strict hardware constraints
- Integrated peripheral hardware including passive buzzers, accelerometers, onboard LEDs, and Bluetooth modules
- Implemented concurrency with locks, 4 original libraries, custom round-robin scheduler, I2C interface

StepperFX (Java)

Jun. 2025– May 2026

- Engineered enterprise-grade GUI application, implementing cipher designed to solve Vigenere vulnerabilities
- Implemented Model-View-Controller architecture, safe multithreading, and 83 automated tests with boundary cases

Geodata Project Team

Aug. 2025 – present

- Organized fundraising campaign for research, compiled database of donors and corporate sponsors
- Designed website under strict deadlines, adding reusable animated JSX components and centralizing CSS stylesheets